Problem 6

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Problem 1 *A company is producing cell phones. The cost of producing x cell phones is given by $C(x)$, defined by*

$$C(x) = -0.01x^2 + 40x + 400, \text{ for } 0 \leq x < 1000.$$

AVERAGE cost of producing the first x cell phones is given by

$$C_{AVG} = \frac{C(x)}{x}$$

If the company has produced x cell phones, the cost of producing one more item is given by

$$\textit{COST of producing one more item} = C(x+1) - C(x)$$

MARGINAL COST is approximation of the cost of producing one more cell phone

$$\textit{MARGINAL COST} = C'(x)$$

(a) *Compute the average cost of the first 300 cellphones that the company produces.*

(b) *Compute the cost of producing one more cell phone, if the company has produced 300 cell phones.*

(c) *Compute the marginal cost, if 300 cell phones have been produced.*

(d) *Why is the marginal cost a good approximation of the cost of producing one more item? Explain!*